WAF++

Architecture Review Report

Architecture Review Report

Infrastructure Security & Compliance Assessment

Checked path	../dummy_code
Report date	2026-03-18 17:51 UTC
Controls loaded	70
Controls run	70
Tool version	WAF++ PASS v0.1.0

15	38	17
PASS	FAIL	SKIP

Executive Summary

Metric	Controls	Checks
✓ Passed	15	104
✗ Failed	38	81
■ Skipped	17	112
Total	70	297

Pillar Breakdown

Pillar	✓ Pass	✗ Fail	■ Skip
Cost	2	6	1
Operations	1	4	5
Performance	2	5	1
Reliability	2	3	5
Security	5	5	3
Sovereign	2	8	0
Sustainability	1	7	2

Controls Overview

All evaluated controls with their overall result. Detailed findings for failed controls follow in the next section.

ID	Title	Pillar	Severity	Status	✓	✗	■
WAF-COST-010	Cost Allocation Tagging Enforced	Cost	HIGH	FAIL	10	4	1
WAF-COST-020	Cost Budgets & Alerting Configured	Cost	HIGH	PASS	2	0	1
WAF-COST-030	Resource Rightsizing & Idle Detection	Cost	MEDIUM	FAIL	0	1	1
WAF-COST-040	Storage & Retention Lifecycle Defined	Cost	HIGH	FAIL	6	1	0
WAF-COST-050	Cost Impact Assessment in Architecture Dec…	Cost	HIGH	SKIP	0	0	1
WAF-COST-060	FinOps Review Cadence	Cost	MEDIUM	FAIL	0	2	0
WAF-COST-070	Observability & Logging Cost Tiers	Cost	MEDIUM	FAIL	9	1	0
WAF-COST-080	Commitment & Reserved Capacity Planning	Cost	MEDIUM	FAIL	0	1	1
WAF-COST-090	Data Transfer & Egress Cost Management	Cost	HIGH	PASS	2	0	0
WAF-OPS-010	CI/CD Pipeline Defined & Automated	Operations	HIGH	SKIP	0	0	3
WAF-OPS-020	Infrastructure as Code Enforced	Operations	HIGH	FAIL	0	2	1
WAF-OPS-030	Observability Stack Configured	Operations	HIGH	FAIL	2	4	1
WAF-OPS-040	Alerting on Symptoms, Not Causes	Operations	HIGH	FAIL	0	1	2
WAF-OPS-050	Change Management & Deployment Risk Assess…	Operations	MEDIUM	SKIP	0	0	2
WAF-OPS-060	Runbook & Operational Documentation Covera…	Operations	MEDIUM	FAIL	0	1	1
WAF-OPS-070	Post-Incident Review Process	Operations	MEDIUM	SKIP	0	0	2
WAF-OPS-080	Feature Flag & Safe Deployment Patterns	Operations	MEDIUM	SKIP	0	0	3
WAF-OPS-090	Configuration Drift Detection & Remediatio…	Operations	HIGH	PASS	2	0	1

WAF-OPS-100	Operational Debt Register & Review	Operations	MEDIUM	SKIP	0	0	2
WAF-PERF-010	Compute Instance Type & Sizing Validated	Performance	HIGH	PASS	1	0	2
WAF-PERF-020	Auto-Scaling Configured & Tested	Performance	HIGH	SKIP	0	0	3
WAF-PERF-030	Caching Strategy Defined & Implemented	Performance	MEDIUM	FAIL	0	1	2
WAF-PERF-040	Database Performance Baseline & Index Stra…	Performance	HIGH	FAIL	0	1	2
WAF-PERF-050	Performance Monitoring & SLO Definition	Performance	HIGH	FAIL	0	1	2
WAF-PERF-070	Network Latency & Topology Optimization	Performance	MEDIUM	PASS	1	0	2
WAF-PERF-080	Serverless & Managed Services for Variable…	Performance	LOW	FAIL	0	1	2
WAF-PERF-090	Storage I/O Performance & Throughput Optim…	Performance	MEDIUM	FAIL	0	1	2
WAF-REL-010	SLA & SLO Definition Documented	Reliability	CRITICAL	FAIL	0	1	2
WAF-REL-020	Health Checks & Readiness Probes Configure…	Reliability	HIGH	SKIP	0	0	3
WAF-REL-030	Multi-AZ High Availability Deployment	Reliability	HIGH	PASS	1	0	3
WAF-REL-040	Backup & Recovery Validation	Reliability	CRITICAL	FAIL	1	1	2
WAF-REL-050	Circuit Breaker & Timeout Configuration	Reliability	HIGH	SKIP	0	0	3
WAF-REL-060	Incident Response & Runbook Readiness	Reliability	HIGH	FAIL	0	1	2
WAF-REL-070	Disaster Recovery Testing	Reliability	HIGH	SKIP	0	0	3
WAF-REL-080	Dependency & Upstream Resilience Managemen…	Reliability	MEDIUM	PASS	1	0	2
WAF-REL-090	Chaos Engineering & Fault Injection	Reliability	MEDIUM	SKIP	0	0	3
WAF-REL-100	Reliability Debt Register & Quarterly Revi…	Reliability	MEDIUM	SKIP	0	0	3
WAF-SEC-010	Identity & Access Management Baseline	Security	CRITICAL	FAIL	0	1	0
WAF-SEC-020	Least Privilege & RBAC Enforcement	Security	CRITICAL	PASS	4	0	0

WAF-SEC-030	Encryption at Rest with Customer-Managed K…	Security	HIGH	FAIL	1	1	0
WAF-SEC-040	Encryption in Transit – TLS Enforcement	Security	HIGH	SKIP	0	0	2
WAF-SEC-050	Network Segmentation & Security Group Hard…	Security	HIGH	PASS	1	0	2
WAF-SEC-060	Secrets Management – No Hardcoded Credenti…	Security	CRITICAL	FAIL	2	2	0
WAF-SEC-070	Vulnerability & Patch Management	Security	MEDIUM	SKIP	0	0	2
WAF-SEC-080	Security Monitoring & Threat Detection	Security	HIGH	PASS	3	0	0
WAF-SEC-090	Policy-as-Code & Compliance Automation	Security	MEDIUM	PASS	1	0	1
WAF-SEC-100	Incident Response Readiness	Security	MEDIUM	FAIL	0	5	1
WAF-SEC-110	Supply Chain Security & SBOM	Security	HIGH	SKIP	0	0	2
WAF-SEC-120	Container & Runtime Security	Security	HIGH	PASS	1	0	1
WAF-SEC-130	Data Classification & Sensitive Data Prote…	Security	HIGH	FAIL	2	6	0
WAF-SOV-010	Data Residency Policy Defined	Sovereign	HIGH	FAIL	4	2	2
WAF-SOV-020	Region Pinning Enforced (IaC)	Sovereign	CRITICAL	PASS	16	0	1
WAF-SOV-030	Backup Location & Retention Controlled	Sovereign	HIGH	FAIL	0	2	4
WAF-SOV-040	Logging & Telemetry Residency Controlled	Sovereign	HIGH	FAIL	3	10	0
WAF-SOV-050	Key Ownership & Management Defined	Sovereign	CRITICAL	FAIL	2	5	1
WAF-SOV-060	Privileged Access Controlled (Separation o…	Sovereign	CRITICAL	FAIL	4	1	3
WAF-SOV-070	Break-Glass Process & Logging	Sovereign	HIGH	PASS	2	0	3
WAF-SOV-080	Dependency & Subprocessor Inventory	Sovereign	MEDIUM	FAIL	1	1	2
WAF-SOV-090	Controlled Egress & Data Exfiltration Guar…	Sovereign	HIGH	FAIL	2	2	2
WAF-SOV-100	Exit Plan & Portability Tested	Sovereign	MEDIUM	FAIL	5	5	1

WAF-SUS-010	Carbon Footprint Measurement & Reporting	Sustainability	HIGH	FAIL	6	1	1
WAF-SUS-020	Energy-Efficient Compute Selection	Sustainability	HIGH	FAIL	1	1	1
WAF-SUS-030	Green Region & Carbon-Aware Workload Place…	Sustainability	MEDIUM	FAIL	0	2	1
WAF-SUS-040	Idle & Underutilized Resource Elimination	Sustainability	HIGH	SKIP	0	0	3
WAF-SUS-050	Storage Lifecycle & Data Minimization	Sustainability	MEDIUM	FAIL	2	1	1
WAF-SUS-060	Workload Scheduling & Time-Shifting	Sustainability	LOW	SKIP	0	0	3
WAF-SUS-070	Sustainable Software Design Standards	Sustainability	MEDIUM	FAIL	0	1	2
WAF-SUS-080	Network & Data Transfer Efficiency	Sustainability	MEDIUM	PASS	1	0	2
WAF-SUS-090	ESG Reporting & Compliance Automation	Sustainability	MEDIUM	FAIL	2	4	1
WAF-SUS-100	Sustainability Debt Register & Quarterly R…	Sustainability	LOW	FAIL	0	2	1

Regulatory Alignment

Overview of how WAF++ controls map to the key regulatory and industry frameworks. Each card shows the compliance posture (PASS / FAIL / SKIP) for all controls that reference the framework.

Framework	Mapped Controls	✓ Pass	✗ Fail	■ Skip
GDPR	25	5	18	2
BSI C5:2020	25	8	14	3
ISO 27001:2022	23	7	13	3
EUCS (ENISA)	23	7	13	3
SOC 2 Type II	5	1	2	2
AWS Well-Architected Framework	27	4	12	11

Framework Cards

Each card shows the framework initials, PASS / FAIL / SKIP counts, and the WAF++ control IDs that reference it. Drop a logo PNG into assets/regulatory/ to replace the initials badge.

G GDPR

✓ 5 | ✗ 18 | ■ 2

WAF-COST-040 ✗ · WAF-COST-070 ✗ · WAF-REL-040 ✗ · WAF-REL-060 ✗ · WAF-REL-070 ■ · WAF-SEC-010 ✗ · WAF-SEC-020 ✓ · WAF-SEC-030 ✗ · WAF-SEC-040 ■ · WAF-SEC-050 ✓ · WAF-SEC-060 ✗ · WAF-SEC-080 ✓ · WAF-SEC-100 ✗ · WAF-SEC-130 ✗ · WAF-SOV-010 ✗ · WAF-SOV-020 ✓ · WAF-SOV-030 ✗ · WAF-SOV-040 ✗ · WAF-SOV-050 ✗ · WAF-SOV-060 ✗ · WAF-SOV-070 ✓ · WAF-SOV-080 ✗ · WAF-SOV-090 ✗ · WAF-SOV-100 ✗ · WAF-SUS-050 ✗

BC BSI C5:2020

✓ 8 | ✗ 14 | ■ 3

WAF-OPS-030 ✗ · WAF-OPS-090 ✓ · WAF-SEC-010 ✗ · WAF-SEC-020 ✓ · WAF-SEC-030 ✗ · WAF-SEC-040 ■ · WAF-SEC-050 ✓ · WAF-SEC-060 ✗ · WAF-SEC-070 ■ · WAF-SEC-080 ✓ · WAF-SEC-090 ✓ · WAF-SEC-100 ✗ · WAF-SEC-110 ■ · WAF-SEC-120 ✓ · WAF-SEC-130 ✗ · WAF-SOV-010 ✗ · WAF-SOV-020 ✓ · WAF-SOV-030 ✗ · WAF-SOV-040 ✗ · WAF-SOV-050 ✗ · WAF-SOV-060 ✗ · WAF-SOV-070 ✓ · WAF-SOV-080 ✗ · WAF-SOV-090 ✗ · WAF-SOV-100 ✗

I2 ISO 27001:2022

✓ 7	✗ 13	■ 3

WAF-SEC-010 ✗ · WAF-SEC-020 ✓ · WAF-SEC-030 ✗ ·
WAF-SEC-040 ■ · WAF-SEC-050 ✓ · WAF-SEC-060 ✗ ·
WAF-SEC-070 ■ · WAF-SEC-080 ✓ · WAF-SEC-090 ✓ ·
WAF-SEC-100 ✗ · WAF-SEC-110 ■ · WAF-SEC-120 ✓ ·
WAF-SEC-130 ✗ · WAF-SOV-010 ✗ · WAF-SOV-020 ✓ ·
WAF-SOV-030 ✗ · WAF-SOV-040 ✗ · WAF-SOV-050 ✗ ·
WAF-SOV-060 ✗ · WAF-SOV-070 ✓ · WAF-SOV-080 ✗ ·
WAF-SOV-090 ✗ · WAF-SOV-100 ✗

E(EUCS (ENISA)

✓ 7	✗ 13	■ 3

WAF-SEC-010 ✗ · WAF-SEC-020 ✓ · WAF-SEC-030 ✗ ·
WAF-SEC-040 ■ · WAF-SEC-050 ✓ · WAF-SEC-060 ✗ ·
WAF-SEC-070 ■ · WAF-SEC-080 ✓ · WAF-SEC-090 ✓ ·
WAF-SEC-100 ✗ · WAF-SEC-110 ■ · WAF-SEC-120 ✓ ·
WAF-SEC-130 ✗ · WAF-SOV-010 ✗ · WAF-SOV-020 ✓ ·
WAF-SOV-030 ✗ · WAF-SOV-040 ✗ · WAF-SOV-050 ✗ ·
WAF-SOV-060 ✗ · WAF-SOV-070 ✓ · WAF-SOV-080 ✗ ·
WAF-SOV-090 ✗ · WAF-SOV-100 ✗

S2T SOC 2 Type II

✓ 1	✗ 2	■ 2

WAF-OPS-030 ✗ · WAF-OPS-050 ■ · WAF-OPS-060 ✗ ·
WAF-OPS-070 ■ · WAF-OPS-090 ✓

AWF AWS Well-Architected Framework

✓ 4	✗ 12	■ 11

WAF-OPS-010 ■ · WAF-OPS-020 ✗ · WAF-OPS-030 ✗ ·
WAF-OPS-040 ✗ · WAF-OPS-050 ■ · WAF-OPS-060 ✗ ·
WAF-OPS-070 ■ · WAF-OPS-080 ■ · WAF-OPS-100 ■ ·
WAF-PERF-010 ✓ · WAF-PERF-020 ■ · WAF-PERF-030 ✗ ·
WAF-PERF-040 ✗ · WAF-PERF-050 ✗ · WAF-PERF-070 ✓ ·
WAF-PERF-080 ✗ · WAF-PERF-090 ✗ · WAF-REL-010 ✗ ·
WAF-REL-020 ■ · WAF-REL-030 ✓ · WAF-REL-040 ✗ ·
WAF-REL-050 ■ · WAF-REL-060 ✗ · WAF-REL-070 ■ ·
WAF-REL-080 ✓ · WAF-REL-090 ■ · WAF-REL-100 ■

Findings

38 control(s) produced findings that require attention. Each finding includes the affected resource, the specific violation, and remediation guidance.

WAF-COST-010 Cost Allocation Tagging Enforced

HIGH

All cloud resources MUST carry mandatory cost allocation tags: cost-center, owner, environment, and workload. These tags are required on all resource types that generate billable costs (compute, storage, database, network, managed services). Tagging MUST be enforced in the CI/CD pipeline; deployments without full tag compliance MUST be blocked or f…

FAIL	`aws_instance.web`	Resource is missing the mandatory 'cost-center' tag.; Resource is missing the mandatory 'owner' tag.
FAIL	`aws_eks_node_group.main`	Resource is missing the mandatory 'workload' tag.
FAIL	`aws_elasticache_cluster.session`	Resource is missing the mandatory 'owner' tag.; Resource is missing the mandatory 'workload' tag.
FAIL	`aws_s3_bucket.logs_raw`	Resource is missing the mandatory 'cost-center' tag.; Resource is missing the mandatory 'owner' tag.; Resource is missing the mandatory 'workload' tag.

→ **Remediation:** Use the shared mandatory-tags module to generate the complete tag map. Add module "tags" { source = "../../modules/mandatory-tags"; cost_center = ...; owner = ...; environment = ...; workload = ...; } Then set tags = module.tags.tags on every resource.

WAF-COST-030 Resource Rightsizing & Idle Detection

MEDIUM

All persistent compute resources (EC2 instances, VMs, RDS instances, Kubernetes node groups) MUST carry a 'rightsizing-reviewed' tag with the date of the last review (ISO 8601, YYYY-MM-DD). Reviews must be performed at least every 90 days. Resources tagged with a review date older than 90 days are considered non-compliant. Idle detection must be co…

FAIL	`aws_instance.web`	EC2 instance is missing the 'rightsizing-reviewed' tag. All compute resources must be actively reviewed.

→ **Remediation:** Add 'rightsizing-reviewed' tag to all EC2 instances and launch templates. Value format: YYYY-MM-DD (ISO 8601 date of last review). Example: rightsizing-reviewed = "2025-03-01" Schedule the next review within 90 days.

WAF-COST-040 Storage & Retention Lifecycle Defined

HIGH

All storage resources (S3 buckets, Azure Storage accounts, GCS buckets) MUST have explicit lifecycle rules configured. All log groups (CloudWatch, Azure Log Analytics, GCP Cloud Logging) MUST have an explicit retention_in_days value that is not 0 (infinite) and not null. No storage or logging resource may be provisioned without an explicit, finite …

FAIL	`aws_cloudwatch_log_group.debug_logs`	retention_in_days must be >= 1. Setting 0 means infinite retention (WAF-COST-040 violation).

→ **Remediation:** Set retention_in_days on all aws_cloudwatch_log_group resources. Use the 4-tier model: operational=30, security-audit=365, compliance=2555 (7yr). Never use 0 (infinite) as this represents unconstrained cost debt.

WAF-COST-060 FinOps Review Cadence

MEDIUM

Monthly engineering-level FinOps reviews MUST take place for all production workloads. Quarterly Architecture Board reviews covering cost debt and strategic cost decisions MUST occur. Review outputs (action items with owner and due date) MUST be documented in writing and accessible in the team's task tracking system. Reviews are mandatory, not opti…

FAIL	`aws_instance.web`	Resource is missing 'last-finops-review' tag. Monthly FinOps reviews required.
FAIL	`aws_eks_cluster.main`	Resource is missing 'last-finops-review' tag. Monthly FinOps reviews required.

→ **Remediation:** After each monthly FinOps review, update the 'last-finops-review' tag on workload resources to the current date. This can be automated in CI as part of the review process. Value format: YYYY-MM-DD.

WAF-COST-070 Observability & Logging Cost Tiers

MEDIUM

Logging and observability resources MUST be organized in explicit retention tiers: Hot (7-30 days), Warm (30-90 days), Cold (90-365 days), Archive (>365 days). CloudWatch Log Groups MUST NOT have retention_in_days = 0 (infinite). Operational log groups MUST have retention_in_days <= 365. DEBUG-level logging in production is prohibited without Archi…

FAIL	`aws_cloudwatch_log_group.debug_logs`	retention_in_days must be >= 1. Value 0 means infinite retention (cost debt).

→ **Remediation:** Set retention_in_days on all aws_cloudwatch_log_group resources. Recommended values by tier: operational=30, access=90, security-audit=365. Compliance-driven long retention: use S3 export + lifecycle instead of CloudWatch.

WAF-COST-080 Commitment & Reserved Capacity Planning

MEDIUM

Compute resources with sustained utilization >= 70% over 30 days MUST be covered by Reserved Instances or Savings Plans. All persistent on-demand compute resources MUST carry a 'capacity-commitment' tag documenting their reservation status and the date of the last commitment review. Spot/Preemptible Instances are the preferred option for variable a…

FAIL	`aws_instance.web`	EC2 instance missing 'capacity-commitment' tag. Reservation status must be declared.; capacity-commitment must be: reserved, savings-plan, spot, on-demand, or reviewed.

→ **Remediation:** Add 'capacity-commitment' tag to all EC2 instances. Value 'reserved' or 'savings-plan' = commitment exists. Value 'on-demand' = no commitment, flagged for review. Value 'reviewed' = reviewed and consciously left on-demand (needs date + justification).

WAF-OPS-020 Infrastructure as Code Enforced

HIGH

All cloud infrastructure MUST be defined as code using an IaC tool (Terraform, Pulumi, AWS CDK, or equivalent). Manual resource creation through cloud consoles MUST be prohibited for all production and staging environments. Infrastructure definitions MUST be version-controlled, reviewed through pull requests, and deployed exclusively through CI/CD …

FAIL	`terraform`	Terraform must configure a remote backend (not local state).
FAIL	`aws_s3_bucket_versioning.data_lake`	S3 state bucket versioning must be Enabled.

→ **Remediation:** Configure the S3 backend in your Terraform root module's terraform block. Create the S3 bucket with versioning enabled and a DynamoDB table for locking.

WAF-OPS-030 Observability Stack Configured

HIGH

Every production workload MUST emit structured logs, expose metrics, and support distributed tracing. An observability stack MUST be configured with centralized log aggregation, a metrics collection system, and distributed tracing. Logs MUST be structured (JSON), include trace IDs and request IDs, and be routed to a centralized platform with define…

FAIL	`aws_cloudwatch_log_group.access_logs`	Log group retention must be at least 30 days.
FAIL	`aws_cloudwatch_log_group.debug_logs`	Log group retention must be at least 30 days.
FAIL	`aws_cloudwatch_log_group.vpc_flow_logs`	Log group retention must be at least 30 days.
FAIL	`aws_lambda_function.event_processor`	Lambda tracing mode must be Active, not PassThrough.

→ **Remediation:** Add retention_in_days to all aws_cloudwatch_log_group resources. Recommended: 90 days for application logs, 365 days for security/audit logs.

WAF-OPS-040 Alerting on Symptoms, Not Causes

HIGH

All production alerting MUST be based on user-visible symptoms (elevated error rates, latency exceeding SLO thresholds, availability degradation) rather than internal causes (CPU usage, memory consumption, disk I/O). Every alert that pages an on-call engineer MUST be actionable and MUST have a linked runbook. Alert fatigue through non-actionable al…

FAIL	`aws_cloudwatch_metric_alarm.high_lambda_errors`	CloudWatch alarm must have at least one alarm_action (SNS topic, Auto Scaling, etc.).

→ **Remediation:** Add alarm_description with symptom description and runbook link. Ensure alarm_actions routes to an SNS topic connected to on-call alerting.

WAF-OPS-060 Runbook & Operational Documentation Coverage

MEDIUM

Every production workload MUST have a runbook covering all known failure modes, routine operational tasks, and escalation paths. Runbooks MUST be version-controlled, regularly reviewed (at minimum quarterly), and accessible to all on-call engineers without authentication barriers at 3 AM. Alert rules that trigger on-call pages MUST link to the rele…

FAIL	`aws_cloudwatch_metric_alarm.high_lambda_errors`	CloudWatch alarm description must include a runbook URL (http:// or https://).

→ **Remediation:** Include a runbook URL in the alarm_description field: alarm_description = "Error rate exceeded. Runbook: https://wiki/runbooks/service-errors"

WAF-PERF-030 Caching Strategy Defined & Implemented

MEDIUM

All workloads with repeated read patterns or high-latency data sources MUST have a caching strategy defined and implemented. The caching strategy MUST specify cache layers (application, distributed cache, CDN), TTL policies, cache invalidation mechanisms, and cache hit rate targets. CDN caching MUST be configured for all public-facing static and dy…

FAIL	`aws_elasticache_cluster.session`	ElastiCache replication group must have automatic_failover_enabled = true for production use.; ElastiCache must have at least 2 cache nodes for redundancy.; ElastiCache must have at-rest encryption enabled.

→ **Remediation:** Set automatic_failover_enabled = true, num_cache_clusters >= 2, and at_rest_encryption_enabled = true. For Redis, use replication_group resource with at least one replica.

WAF-PERF-040 Database Performance Baseline & Index Strategy

HIGH

All relational and document databases used in production MUST have a documented performance baseline including query execution times, I/O throughput, and connection pool utilization. An index strategy MUST be defined and maintained for all tables with significant read/write loads. Slow query logging MUST be enabled and reviewed regularly. Database …

FAIL	`aws_db_instance.main`	RDS instance must export logs to CloudWatch (slowquery, error, general).; Enhanced Monitoring must be enabled (monitoring_interval >= 1).

→ **Remediation:** Set performance_insights_enabled = true, monitoring_interval = 60 (or 1 for high-granularity). Set enabled_cloudwatch_logs_exports = ["slowquery", "error"] for MySQL or ["postgresql"] for PostgreSQL.

WAF-PERF-050 Performance Monitoring & SLO Definition

HIGH

All production services MUST have Service Level Objectives (SLOs) defined for performance-relevant metrics: latency (P95 and P99), error rate, and availability. Service Level Indicators (SLIs) MUST be instrumented and measured continuously. Performance dashboards MUST be available for all production services showing current SLO compliance and histo…

FAIL	`aws_cloudwatch_metric_alarm.high_lambda_errors`	CloudWatch alarm must have at least one action (SNS topic) configured.

→ **Remediation:** Ensure CloudWatch alarms have alarm_actions pointing to SNS topics for notification. Set evaluation_periods >= 2 to reduce false positives. Add alarm_description with runbook link.

WAF-PERF-080 Serverless & Managed Services for Variable Load

LOW

Workloads with highly variable, spiky, or unpredictable traffic patterns SHOULD be evaluated for serverless or fully managed compute architectures. Where serverless is appropriate, Lambda, Cloud Functions, Azure Functions, or managed container services (Fargate, Cloud Run) MUST be configured with appropriate memory, timeout, concurrency, and cold-s…

FAIL	`aws_lambda_function.event_processor`	Lambda memory_size must be explicitly set >= 256MB for production functions (default 128MB is rarely optimal).; Lambda timeout must be explicitly set (not default 3s) and sized appropriately.; Lambda reserved_concurrent_executions should be configured to control account concurrency usage.

→ **Remediation:** Run AWS Lambda Power Tuning to find optimal memory allocation. Set timeout = 2x measured P99 execution time. Set reserved_concurrent_executions to limit blast radius on account concurrency.

WAF-PERF-090 Storage I/O Performance & Throughput Optimization

MEDIUM

All block storage (EBS, Azure Managed Disks, GCP Persistent Disk) and object storage used by production workloads MUST be configured with appropriate I/O performance settings (IOPS, throughput, storage class). Storage volumes MUST use the most cost-efficient performance tier for their workload. EBS volumes MUST use gp3 instead of gp2 for new deploy…

FAIL	`aws_db_instance.main`	EBS volume must not use gp2 type. Migrate to gp3 for consistent performance and lower cost.; EBS volume type must be explicitly set to a supported current type.

→ **Remediation:** Change volume_type from "gp2" to "gp3". For databases requiring higher IOPS, set explicit iops and throughput values on gp3 (max 16000 IOPS, 1000 MB/s). Migration from gp2 to gp3 is online and non-disruptive.

WAF-REL-010 SLA & SLO Definition Documented

CRITICAL

Every production workload MUST have documented Service Level Objectives (SLOs) and Service Level Agreements (SLAs). SLOs must define availability, latency, error rate, and throughput targets with explicit measurement windows. SLAs must reference SLOs and define consequences of non-compliance. Both documents must be version-controlled, reviewed quar…

FAIL	`aws_cloudwatch_metric_alarm.high_lambda_errors`	CloudWatch alarm must have at least one action configured.

→ **Remediation:** Define a CloudWatch alarm that monitors the primary SLO metric (e.g., error rate, latency). Configure alarm_actions to notify SNS topics used by on-call rotation.

WAF-REL-040 Backup & Recovery Validation

CRITICAL

All production databases, object stores, and stateful services MUST have automated backup configurations with defined retention periods. Backup configurations MUST specify RPO-aligned retention. Recovery procedures MUST be tested at minimum quarterly through documented restore exercises. Backups MUST be stored in a separate account, region, or vaul…

FAIL	`aws_s3_bucket_versioning.data_lake`	S3 bucket versioning must be Enabled for production data buckets.

→ **Remediation:** Add an aws_s3_bucket_versioning resource referencing the bucket with versioning_configuration { status = "Enabled" }.

WAF-REL-060 Incident Response & Runbook Readiness

HIGH

Every production workload MUST have a documented Incident Response (IR) plan with defined severity levels, escalation paths, and on-call rotations. Runbooks MUST exist for all critical alerts and MUST be version-controlled. Runbooks MUST be linked directly from alert notifications. Post-incident reviews MUST be conducted within 5 business days and …

FAIL	`aws_cloudwatch_metric_alarm.high_lambda_errors`	CloudWatch alarm must have alarm_actions configured for on-call notification.; CloudWatch alarm must have ok_actions to notify when alert resolves.

→ **Remediation:** Add alarm_actions and ok_actions pointing to an SNS topic. The SNS topic should be subscribed to by PagerDuty, OpsGenie, or email for on-call notification.

WAF-SEC-010 Identity & Access Management Baseline

CRITICAL

All AWS accounts MUST enforce MFA for all IAM users with console access. Root account usage MUST be prohibited for day-to-day operations. IAM password policy MUST meet minimum security requirements: minimum length of 14 characters, complexity requirements, password reuse prevention of at least 5 prior passwords, and maximum password age of 90 days.…

FAIL	`aws_iam_account_password_policy.main`	IAM password policy must require a minimum of 14 characters.; IAM password policy must prevent reuse of at least 5 prior passwords.; IAM password policy must set maximum password age to 90 days or less.

→ **Remediation:** Define an aws_iam_account_password_policy resource with all required attributes set. Ensure minimum_password_length >= 14, all require_* flags are true, password_reuse_prevention >= 5, and max_password_age <= 90.

WAF-SEC-030 Encryption at Rest with Customer-Managed Keys

HIGH

All data stores containing sensitive or classified data MUST be encrypted at rest using customer-managed keys (CMK) stored in AWS KMS. Automatic key rotation MUST be enabled for all CMKs. Encryption using only provider-managed keys (SSE-S3/AES256) is NOT sufficient for sensitive, PII, financial, or regulated data. KMS key deletion windows MUST be s…

FAIL	`aws_kms_key.main`	KMS key deletion window must be at least 7 days to allow recovery from accidental deletion.

→ **Remediation:** Set enable_key_rotation = true on all aws_kms_key resources. Set deletion_window_in_days to at least 7 (recommend 30 for production keys). Note: key rotation does not apply to asymmetric keys or keys in external key stores; those require separate rotation procedures.

WAF-SEC-060 Secrets Management – No Hardcoded Credentials

CRITICAL

All secrets, API keys, passwords, database credentials, and certificates MUST be stored in a dedicated secrets management solution (AWS Secrets Manager or HashiCorp Vault). Secrets MUST NOT be hardcoded in IaC code, Terraform variable files, environment variables in task definitions, source code, or CI/CD pipeline configuration. Secrets Manager sec…

FAIL	`aws_secretsmanager_secret.db_password`	Secrets Manager secret recovery window must be at least 7 days to allow recovery from accidental deletion.
FAIL	`aws_secretsmanager_secret.api_key`	Secrets Manager secret recovery window must be at least 7 days to allow recovery from accidental deletion.

→ **Remediation:** Add name and recovery_window_in_days = 30 to all aws_secretsmanager_secret resources. Do not set force_overwrite_replica_secret = true or recovery_window_in_days = 0 unless in a controlled testing environment.

WAF-SEC-100 Incident Response Readiness

MEDIUM

An incident response (IR) plan MUST exist, be documented, version-controlled, and tested at least annually via tabletop exercises or simulated incident drills. SNS topics or equivalent alerting channels MUST be configured to receive security findings from GuardDuty, Security Hub, and CloudTrail metric alarms. Security-relevant CloudWatch log groups…

FAIL	`aws_cloudwatch_log_group.cloudtrail`	Security-relevant log groups MUST retain logs for at least 365 days to support forensic investigation and regulatory compliance.
FAIL	`aws_cloudwatch_log_group.app_logs`	Security-relevant log groups MUST retain logs for at least 365 days to support forensic investigation and regulatory compliance.
FAIL	`aws_cloudwatch_log_group.access_logs`	Security-relevant log groups MUST retain logs for at least 365 days to support forensic investigation and regulatory compliance.
FAIL	`aws_cloudwatch_log_group.debug_logs`	Security-relevant log groups MUST retain logs for at least 365 days to support forensic investigation and regulatory compliance.
FAIL	`aws_cloudwatch_log_group.vpc_flow_logs`	Security-relevant log groups MUST retain logs for at least 365 days to support forensic investigation and regulatory compliance.

→ **Remediation:** Set retention_in_days = 365 (or higher) on all security-relevant aws_cloudwatch_log_group resources. For CloudTrail and VPC Flow Logs with higher volume, consider using S3-based retention with lifecycle policies (Standard for 12 months, Glacier for 7 years) rather than CloudWatch to optimize cost.

WAF-SEC-130 Data Classification & Sensitive Data Protection

HIGH

All data-storing resources (S3 buckets, RDS instances, DynamoDB tables, ElastiCache clusters) MUST carry a 'data-class' tag indicating the sensitivity level (e.g., public, internal, confidential, pii, health, financial). Resources classified as 'pii', 'health', or 'financial' MUST be encrypted with a customer-managed KMS key (CMK). Database instanc…

FAIL	`aws_db_instance.main`	Data-storing resource MUST carry a 'data-class' tag (e.g., public, internal, confidential, pii, health, financial).
FAIL	`aws_elasticache_cluster.session`	Data-storing resource MUST carry a 'data-class' tag (e.g., public, internal, confidential, pii, health, financial).
FAIL	`aws_s3_bucket.data_lake`	Data-storing resource MUST carry a 'data-class' tag (e.g., public, internal, confidential, pii, health, financial).
FAIL	`aws_s3_bucket.logs_raw`	Data-storing resource MUST carry a 'data-class' tag (e.g., public, internal, confidential, pii, health, financial).
FAIL	`aws_s3_bucket.backups`	Data-storing resource MUST carry a 'data-class' tag (e.g., public, internal, confidential, pii, health, financial).
FAIL	`aws_s3_bucket.cloudtrail`	Data-storing resource MUST carry a 'data-class' tag (e.g., public, internal, confidential, pii, health, financial).

→ **Remediation:** Add a 'data-class' tag to all data-storing resources with a value from the approved classification taxonomy (e.g., "public", "internal", "confidential", "pii", "health", "financial"). Enforce mandatory tagging via an AWS Config rule (required-tags) or an SCP denying CreateBucket, CreateDBInstance, CreateTable API calls without the required tag. Implement a tagging standard in your Terraform module library to make compliant resource creation the default path.

WAF-SOV-010 Data Residency Policy Defined

HIGH

A documented data-residency policy MUST exist that explicitly defines allowed regions, data classifications, and storage rules for all data types: primary data, replicas, backups, logs, traces, metrics, and metadata. The policy MUST be version-controlled and technically reflected in IaC defaults and variable validation.

FAIL	`aws_elasticache_cluster.session`	Resource is missing the 'data-residency' tag required by sovereignty policy.; Resource is missing the 'data-class' tag required by data classification policy.
FAIL	`aws_s3_bucket.logs_raw`	Resource is missing the 'data-residency' tag required by sovereignty policy.; Resource is missing the 'data-class' tag required by data classification policy.

→ **Remediation:** Add 'data-residency' and 'data-class' tags to all data-storing resources. Acceptable values for 'data-residency': eu-only, de-only, ch-only, at-only, global-approved. Acceptable values for 'data-class': pii, health, financial, operational, public, restricted.

WAF-SOV-030 Backup Location & Retention Controlled

HIGH

Backup policies MUST explicitly define region/location constraints, minimum retention periods, and restore-test schedules. Cross-region replication MUST be explicitly configured and documented with an approved sovereign destination. Backup storage MUST remain within the same jurisdictional boundary as primary data unless explicitly approved and doc…

FAIL	`aws_db_instance.main`	RDS backup_retention_period must be >= 7 days for sovereign compliance.
FAIL	`aws_s3_bucket_versioning.data_lake`	S3 bucket versioning must be Enabled for backup/sovereign data stores.

→ **Remediation:** Set backup_retention_period to at least 7 (days) on all production RDS instances and 30 days for instances processing PII or financial data.

WAF-SOV-040 Logging & Telemetry Residency Controlled

HIGH

All observability data—logs, traces, and metrics—MUST be stored in approved sovereign regions. Exports to external platforms (SaaS observability, third-party SIEM) MUST be explicitly approved, documented, and controlled. Log retention periods MUST be defined and enforced. VPC flow logs MUST be enabled to support forensic investigation within sovere…

FAIL	`aws_cloudwatch_log_group.cloudtrail`	Log group retention must be at least 30 days.
FAIL	`aws_cloudwatch_log_group.app_logs`	Log group retention must be at least 30 days.
FAIL	`aws_cloudwatch_log_group.access_logs`	Log group retention must be at least 30 days.
FAIL	`aws_cloudwatch_log_group.debug_logs`	Log group retention must be at least 30 days.
FAIL	`aws_cloudwatch_log_group.vpc_flow_logs`	Log group retention must be at least 30 days.
FAIL	`aws_cloudwatch_log_group.cloudtrail`	Audit/security log group should be encrypted with a customer-managed KMS key.
FAIL	`aws_cloudwatch_log_group.app_logs`	Audit/security log group should be encrypted with a customer-managed KMS key.
FAIL	`aws_cloudwatch_log_group.access_logs`	Audit/security log group should be encrypted with a customer-managed KMS key.
FAIL	`aws_cloudwatch_log_group.debug_logs`	Audit/security log group should be encrypted with a customer-managed KMS key.
FAIL	`aws_cloudwatch_log_group.vpc_flow_logs`	Audit/security log group should be encrypted with a customer-managed KMS key.

→ **Remediation:** Set retention_in_days on all CloudWatch log groups. Use 90 days minimum for audit and security logs, 30 days minimum for application logs.

WAF-SOV-050 Key Ownership & Management Defined

CRITICAL

For all data classes, the key ownership model MUST be explicitly defined and technically enforced: Provider-Managed Keys (PMK), Customer-Managed Keys (CMK), Bring Your Own Key (BYOK), or Hold Your Own Key (HYOK). Key rotation and deletion MUST be governed by documented procedures. All data at rest MUST be encrypted; encryption with default provider…

FAIL	`aws_kms_key.main`	KMS key deletion_window_in_days must be >= 14 days.
FAIL	`aws_s3_bucket_server_side_encryption_configuration.data_lake`	S3 bucket must use 'aws:kms' encryption, not 'AES256' (provider-managed keys).; S3 KMS encryption must reference a specific CMK, not the default AWS managed key.
FAIL	`aws_s3_bucket_server_side_encryption_configuration.logs_raw`	S3 bucket must use 'aws:kms' encryption, not 'AES256' (provider-managed keys).; S3 KMS encryption must reference a specific CMK, not the default AWS managed key.

FAIL	`aws_s3_bucket_server_side_encryption_configuration.backups`	S3 bucket must use 'aws:kms' encryption, not 'AES256' (provider-managed keys).; S3 KMS encryption must reference a specific CMK, not the default AWS managed key.
FAIL	`aws_instance.web`	EBS volume must have encrypted = true.

→ **Remediation:** Set deletion_window_in_days >= 14 on all KMS keys.

WAF-SOV-060 Privileged Access Controlled (Separation of Duties) CRITICAL

Privileged roles MUST be minimized, time-bound, and subject to regular access reviews. Separation of Duties MUST be enforced: no single principal should be able to create infrastructure, approve deployments, AND manage encryption keys simultaneously. Wildcard IAM permissions ('*' on '*') are forbidden. Administrative access MUST use JIT (Just-In-Ti…

FAIL	`aws_iam_account_password_policy.main`	IAM password minimum length must be >= 14 characters.; IAM password policy must prevent reuse of at least 12 previous passwords.; IAM password max age must be <= 90 days.

→ **Remediation:** Configure an aws_iam_account_password_policy resource with strong password requirements.

WAF-SOV-080 Dependency & Subprocessor Inventory MEDIUM

An inventory of all dependencies and subprocessors MUST be maintained: managed cloud services, third-party SaaS tools, CI/CD platforms, observability providers, and data processors. Changes to this inventory MUST be version-controlled, reviewed, and approved. Provider version constraints in Terraform MUST be pinned to prevent silent upgrades. Modul…

FAIL	`terraform`	Provider version constraint is too broad (>= 0.0.0). Use a tightly bounded constraint.

→ **Remediation:** Add version constraints to all required_providers using the ~> operator for compatible version pinning (e.g., ~> 5.0 for AWS provider v5.x).

WAF-SOV-090 Controlled Egress & Data Exfiltration Guardrails HIGH

Outbound network traffic MUST be controlled using a default-deny or strict allow-list policy. Security groups must not permit unrestricted egress (0.0.0.0/0) without explicit justification. VPC endpoints MUST be used for access to cloud provider APIs to keep traffic on the sovereign network. Data exfiltration detection MUST be established to alert …

FAIL	`aws_security_group.internal_app`	Security group allows unrestricted outbound access to 0.0.0.0/0. Restrict to specific destinations.; Security group allows unrestricted outbound access to ::/0 (all IPv6).
FAIL	`aws_security_group.public_alb`	Security group allows unrestricted outbound access to 0.0.0.0/0. Restrict to specific destinations.; Security group allows unrestricted outbound access to ::/0 (all IPv6).

→ **Remediation:** Replace 0.0.0.0/0 egress rules with specific CIDR ranges or prefix lists. Use VPC endpoints for cloud service access to eliminate the need for internet egress.

WAF-SOV-100 Exit Plan & Portability Tested

MEDIUM

A documented exit plan MUST exist covering data export procedures, IaC portability strategy, dependency replacement mapping, timeline, and cost estimates. Exit/restore drills MUST be performed at least annually to validate that the exit plan is executable. IaC must use open standards and provider-agnostic patterns where feasible to reduce proprieta…

FAIL	`aws_s3_bucket_versioning.data_lake`	S3 bucket versioning must be Enabled for sovereign data portability.
FAIL	`aws_s3_bucket.data_lake`	Resource is missing 'portability-class' tag needed for exit orchestration.
FAIL	`aws_s3_bucket.logs_raw`	Resource is missing 'portability-class' tag needed for exit orchestration.
FAIL	`aws_s3_bucket.backups`	Resource is missing 'portability-class' tag needed for exit orchestration.
FAIL	`aws_s3_bucket.cloudtrail`	Resource is missing 'portability-class' tag needed for exit orchestration.

→ **Remediation:** Enable versioning on S3 buckets containing sovereign data.

WAF-SUS-010 Carbon Footprint Measurement & Reporting

HIGH

Organizations MUST measure and report the carbon footprint of their cloud workloads. Cloud-provider carbon footprint tools (AWS Customer Carbon Footprint Tool, Azure Emissions Impact Dashboard, GCP Carbon Footprint) MUST be enabled and queried at least monthly. Emissions data MUST be exported, tagged by workload, and retained for ESG reporting. Org…

FAIL	`aws_cloudwatch_log_group.debug_logs`	Retention must be set to at least 1 day; recommended 90-365 days for carbon audit data.

→ **Remediation:** Set retention_in_days on all CloudWatch log groups. For carbon footprint audit logs, use 365 days to satisfy CSRD 7-year record retention. Use lifecycle policies to transition older logs to S3 Glacier for cost and energy efficiency.

WAF-SUS-020 Energy-Efficient Compute Selection

HIGH

All compute resources MUST prefer energy-efficient processor architectures (ARM64/Graviton for AWS, Ampere Altra for Azure, T2A for GCP) over equivalent x86 instance families. AWS Lambda functions MUST use arm64 architecture unless the workload has a verified incompatibility. EC2 instances MUST NOT use previous-generation families (t2, m4, c4, r4) …

FAIL	`aws_lambda_function.event_processor`	Lambda function uses x86_64 architecture; switch to arm64 for 20% energy reduction.

→ **Remediation:** Set architectures = ["arm64"] on all Lambda functions. Verify that all Lambda layers and dependencies support arm64. Most Python, Node.js, Java and Go runtimes are fully supported. For container image functions, rebuild the image for linux/arm64 platform.

WAF-SUS-030 Green Region & Carbon-Aware Workload Placement

MEDIUM

Workloads without strict data-residency requirements MUST consider carbon intensity of cloud regions in placement decisions. When architectural and regulatory constraints allow, workloads SHOULD be placed in regions with high renewable energy mix or low grid carbon intensity. Region selection decisions MUST document sustainability considerations al…

FAIL	`provider.aws`	AWS region has a high carbon intensity. Consider us-west-2, eu-north-1, eu-west-1 for lower emissions.
FAIL	`provider.aws`	GCP region has elevated carbon intensity. Consider europe-north1, europe-west1, or us-west1.

→ **Remediation:** If workloads can be placed in a lower-carbon region without violating data residency, latency, or compliance requirements, prefer: us-west-2 (Oregon, ~89% renewable), eu-north-1 (Stockholm, ~95% renewable), eu-west-1 (Ireland, ~80% renewable). If high-carbon region is required, document the justification in an ADR and offset through RECs for that region's workload footprint.

WAF-SUS-050 Storage Lifecycle & Data Minimization

MEDIUM

All storage resources (S3 buckets, EBS volumes, Azure Blob Storage, GCS buckets) MUST have lifecycle policies defined. Data MUST be automatically tiered to cost- and energy-efficient cold storage after defined activity thresholds. Log data MUST have retention limits. Temporary data (build artifacts, test data, cache) MUST have explicit expiration. …

FAIL	`aws_instance.web`	Root EBS volume should be set to delete on termination to prevent orphaned volumes.

→ **Remediation:** Set delete_on_termination = true on root_block_device for EC2 instances unless you have an explicit requirement to retain root volumes. For data volumes, use separate aws_ebs_volume resources with lifecycle management.

WAF-SUS-070 Sustainable Software Design Standards

MEDIUM

Software MUST be designed with energy efficiency as an explicit quality attribute. Architecture Decision Records MUST include energy efficiency considerations for algorithm and data structure choices. Dependencies MUST be reviewed for efficiency: unused libraries, heavy frameworks, and polling patterns MUST be replaced with lightweight alternatives…

FAIL	`aws_lambda_function.event_processor`	No layers attribute found – this check passes; single-package deployments are preferred.

→ **Remediation:** Review Lambda layer usage. Consolidate multiple utility layers into a single optimized layer. Use dependency management (requirements.txt, package.json) to control package size. Consider Lambda container images for complex dependency trees with multi-arch builds.

WAF-SUS-090 ESG Reporting & Compliance Automation

MEDIUM

Organizations subject to CSRD (or voluntarily reporting ESG) MUST have documented processes for collecting, aggregating, and reporting cloud IT emissions as part of Scope 3 GHG inventory. Cloud IT emissions data MUST be exported from cloud provider tools at least quarterly. CSRD-scope organizations MUST map cloud emissions to ESRS E1-6 reporting re…

FAIL	`aws_s3_bucket.data_lake`	ESG/sustainability S3 buckets should have a 'purpose' tag for audit identification.
FAIL	`aws_s3_bucket.logs_raw`	ESG/sustainability S3 buckets should have a 'purpose' tag for audit identification.
FAIL	`aws_s3_bucket.backups`	ESG/sustainability S3 buckets should have a 'purpose' tag for audit identification.

FAIL	`aws_s3_bucket.cloudtrail`	ESG/sustainability S3 buckets should have a 'purpose' tag for audit identification.

→ **Remediation:** Create a dedicated S3 bucket for sustainability data exports. Tag it with purpose = "esg-reporting" and configure lifecycle policies for 7-year retention (CSRD requires 7 years for sustainability disclosures). Enable versioning and access logging for audit trail completeness.

WAF-SUS-100 Sustainability Debt Register & Quarterly Review

LOW

Organizations MUST maintain a Sustainability Debt Register documenting all known gaps between current sustainability posture and target state across all WAF-SUS controls. The register MUST be reviewed and updated at least quarterly. Each entry MUST include: estimated carbon impact, remediation priority, owner, and target resolution date. Sustainabi…

FAIL	`aws_instance.web`	Resource has no sustainability-reviewed tag – add date of last sustainability review.
FAIL	`aws_eks_cluster.main`	Resource has no sustainability-reviewed tag – add date of last sustainability review.

→ **Remediation:** Add sustainability-reviewed = "YYYY-MM-DD" tag to all major compute resources. Set up quarterly reminder to update this tag after sustainability review. Use AWS Config custom rule to alert when sustainability-reviewed tag is older than 180 days.

Passed Controls

Control ID	Title	Checks passed
WAF-COST-020	Cost Budgets & Alerting Configured	2
WAF-COST-090	Data Transfer & Egress Cost Management	2
WAF-OPS-090	Configuration Drift Detection & Remediation	2
WAF-PERF-010	Compute Instance Type & Sizing Validated	1
WAF-PERF-070	Network Latency & Topology Optimization	1
WAF-REL-030	Multi-AZ High Availability Deployment	1
WAF-REL-080	Dependency & Upstream Resilience Management	1
WAF-SEC-020	Least Privilege & RBAC Enforcement	4
WAF-SEC-050	Network Segmentation & Security Group Hardening	1
WAF-SEC-080	Security Monitoring & Threat Detection	3
WAF-SEC-090	Policy-as-Code & Compliance Automation	1
WAF-SEC-120	Container & Runtime Security	1
WAF-SOV-020	Region Pinning Enforced (IaC)	16
WAF-SOV-070	Break-Glass Process & Logging	2
WAF-SUS-080	Network & Data Transfer Efficiency	1

Skipped Controls

The following controls were skipped — either no matching Terraform resources were found, or all assertions use operators not supported in automated evaluation.

Control ID	Title	Reason
WAF-COST-050	Cost Impact Assessment in Architecture Decisi…	No matching resources found in Terraform configuration.
WAF-OPS-010	CI/CD Pipeline Defined & Automated	No matching resources found in Terraform configuration.
WAF-OPS-050	Change Management & Deployment Risk Assessmen…	No matching resources found in Terraform configuration.
WAF-OPS-070	Post-Incident Review Process	No matching resources found in Terraform configuration.
WAF-OPS-080	Feature Flag & Safe Deployment Patterns	No matching resources found in Terraform configuration.
WAF-OPS-100	Operational Debt Register & Review	No matching resources found in Terraform configuration.

WAF-PERF-020	Auto-Scaling Configured & Tested	No matching resources found in Terraform configuration.
WAF-REL-020	Health Checks & Readiness Probes Configured	No matching resources found in Terraform configuration.
WAF-REL-050	Circuit Breaker & Timeout Configuration	No matching resources found in Terraform configuration.
WAF-REL-070	Disaster Recovery Testing	No matching resources found in Terraform configuration.
WAF-REL-090	Chaos Engineering & Fault Injection	No matching resources found in Terraform configuration.
WAF-REL-100	Reliability Debt Register & Quarterly Review	No matching resources found in Terraform configuration.
WAF-SEC-040	Encryption in Transit – TLS Enforcement	No matching resources found in Terraform configuration.
WAF-SEC-070	Vulnerability & Patch Management	No matching resources found in Terraform configuration.
WAF-SEC-110	Supply Chain Security & SBOM	No matching resources found in Terraform configuration.
WAF-SUS-040	Idle & Underutilized Resource Elimination	No matching resources found in Terraform configuration.
WAF-SUS-060	Workload Scheduling & Time-Shifting	No matching resources found in Terraform configuration.